Muthya Varshith Vankamamidi

Senior Software Engineer | Agentic AI, RAG & LLM Systems

Hyderabad, India | me@iamvarshith.dev | linkedin.com/in/muthyavarshith | github.com/iamvarshith | iamvarshith.dev

SUMMARY

Senior Software Engineer with 6 years of experience building agentic AI and developer platforms at scale. Lead engineer of Workato's AgentX Support; built AI agents, RAG pipelines, 99.9998%-accurate multilingual PII detection and 10K+ RPS event-driven pipelines for customer support at 1M+ conversations a day, plus a Zendesk Copilot generating $500K+ ARR. Previously built HackerRank's event-driven testing and CI/CD platform for ~300 engineers, cutting release QA from a week to hours.

EXPERIENCE

Workato — Remote, India

Senior Software Engineer, Agentic AI — Mar 2026 – Present
Software Engineer — Dec 2024 – Feb 2026
Software Engineer, DeepConverse (acquired by Workato) — Sep 2024 – Dec 2024

- Lead engineer and co-architect (with the VP of Engineering) of **AgentX Support**, Workato's agentic support platform on recipes, an AI agent framework and connector-backed skills; owned infrastructure and engineering, shipped in six months.
- Designed the flow engine between the agent framework and decision trees that hands conversations to human agents, including a Salesforce MIAW handover service; built search and agent analytics on Kafka, ClickHouse and Holistics.
- Built the RAG knowledge pipeline: custom crawlers, Apache Airflow-orchestrated chunking across parallel workers, Nomic Embed v1.5 embeddings and Elasticsearch vector search.
- Built a multilingual PII detection service for 7 Asian languages (including Japanese, Korean and Chinese) on a support platform handling 1M conversations/day per customer. LoRA fine-tuned classifiers reached 99.89% accuracy; regex and rule-based layers raised it to **99.9998%** across 2M conversations, on a multi-million-dollar project delivered by five engineers.
- Built an event-driven voice and chat data pipeline (pre-processing, PII redaction, post-processing, summarization, per-customer tasks) across 9 services, each sustaining **10,000+ RPS**.
- Built the Zendesk Copilot, which summarizes tickets, retrieves relevant knowledge, finds past tickets with similar resolutions and flags live incidents, generating **$500K+ ARR**.
- Migrated the legacy DeepConverse platform into Workato with Terraform (32 microservices, ~1M conversations/day, MongoDB, PostgreSQL, Elasticsearch, Redshift): first onto AWS EKS, then onto Workato's Kubernetes clusters, handing infrastructure over to the infra team.
- Integrated MCP servers and Workato Genies, led guardrailed LLM model upgrades that significantly cut model costs, and resolved customer and security issues for legacy accounts.

HackerRank — Bengaluru, India

Software Engineer II — Jul 2023 – Sep 2024
Software Engineer — Jun 2022 – Jul 2023
Software Engineer Intern — Jan 2022 – Jun 2022

- Architected an event-driven regression testing platform (Terraform, RabbitMQ, Argo CD, GitHub Actions) that provisions ephemeral environments with distributed databases per change and runs only impacted tests in parallel; cut major-release QA from **~1 week to hours**, test automation time by 60% and flaky tests by 38%.
- Built Phoenix, a self-healing Selenium UI testing framework that automatically repaired **85%** of broken tests via DOM-similarity matching; later extended with LLMs to 98% and still in production.
- Built Slack ChatOps bots and Terraform infrastructure-as-code tooling used by **~300 engineers** across 8 products to deploy, test and re-run pipelines without per-environment VPNs, reducing environment re-creation from days to minutes.
- Introduced Elasticsearch and Kibana logging and Slack failure alerting for test pipelines; received the Kudos Award for a video-recording framework that generated data for ML plagiarism detection.

Bar-At Technologies – Full Stack Developer, Hyderabad — Feb 2021 – Sep 2021

- Built legal search over past civil court rulings (Elasticsearch) on a cell-based serverless AWS backend: Lambda, DynamoDB.

Freelance – Software Developer (Python, Flask, Django) — Jan 2020 – Feb 2021

SKILLS

AI & LLMs: Agentic AI, AI Agents, RAG, LLM Guardrails, MCP, LoRA Fine-tuning, PII Detection, Embeddings (Nomic), Vector Search, Prompt Engineering, OpenAI
Languages & Frameworks: Python, Ruby, SQL, JavaScript, Shell, FastAPI, Django, Flask, REST APIs, Selenium
Data: Elasticsearch, Kafka, ClickHouse, PostgreSQL, MongoDB, Amazon Redshift, MySQL, Apache Airflow, Holistics, Kibana
Cloud & DevOps: AWS, EKS, Kubernetes, Docker, Terraform, Argo CD, GitHub Actions, Jenkins, RabbitMQ, CI/CD, Event-Driven Architecture, Microservices
Platforms: Workato (Recipes, Connectors, Genies), Zendesk, Salesforce MIAW, Slack API

EDUCATION

B.Tech, Computer Science and Engineering – NIIT University, Neemrana — 2018 – 2022

CERTIFICATIONS & AWARDS

Kudos Award, HackerRank Employee Awards | HackerRank Certified: REST API (Intermediate), Python (Basic), CSS | Tech Team Lead, siNUsoid v4 tech fest, NIIT University